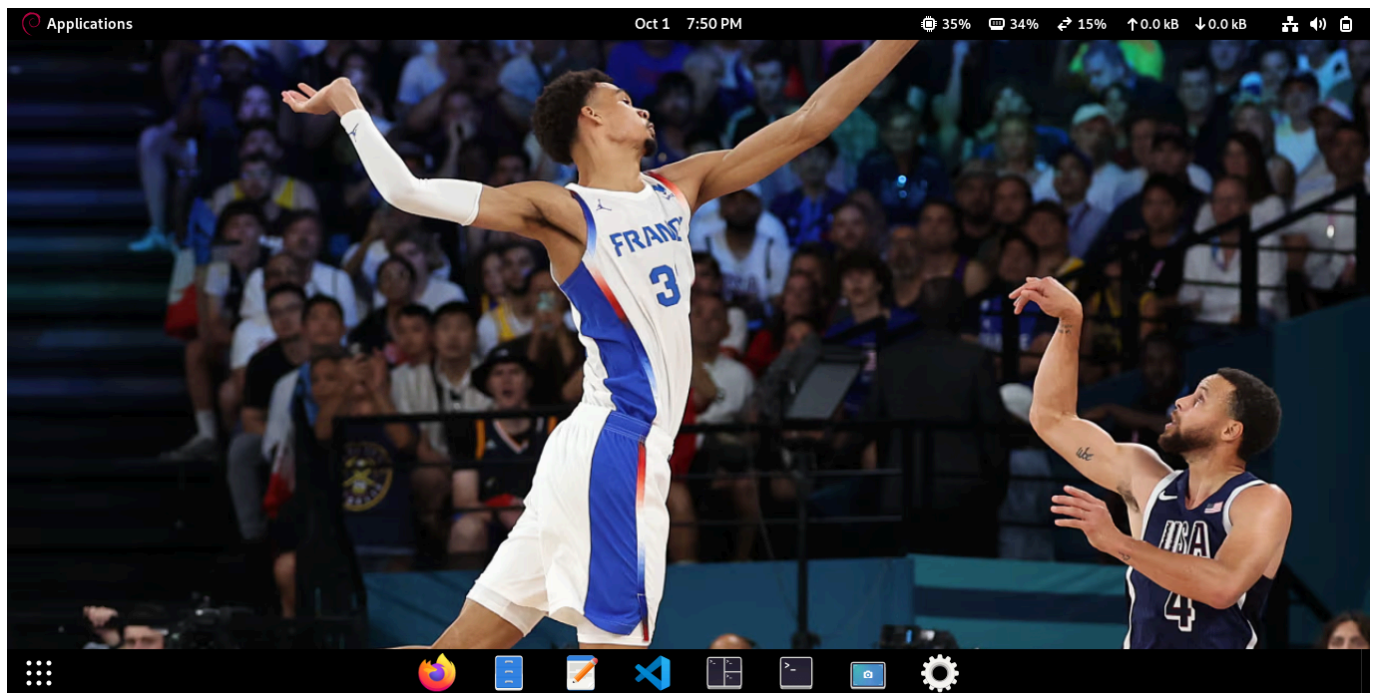
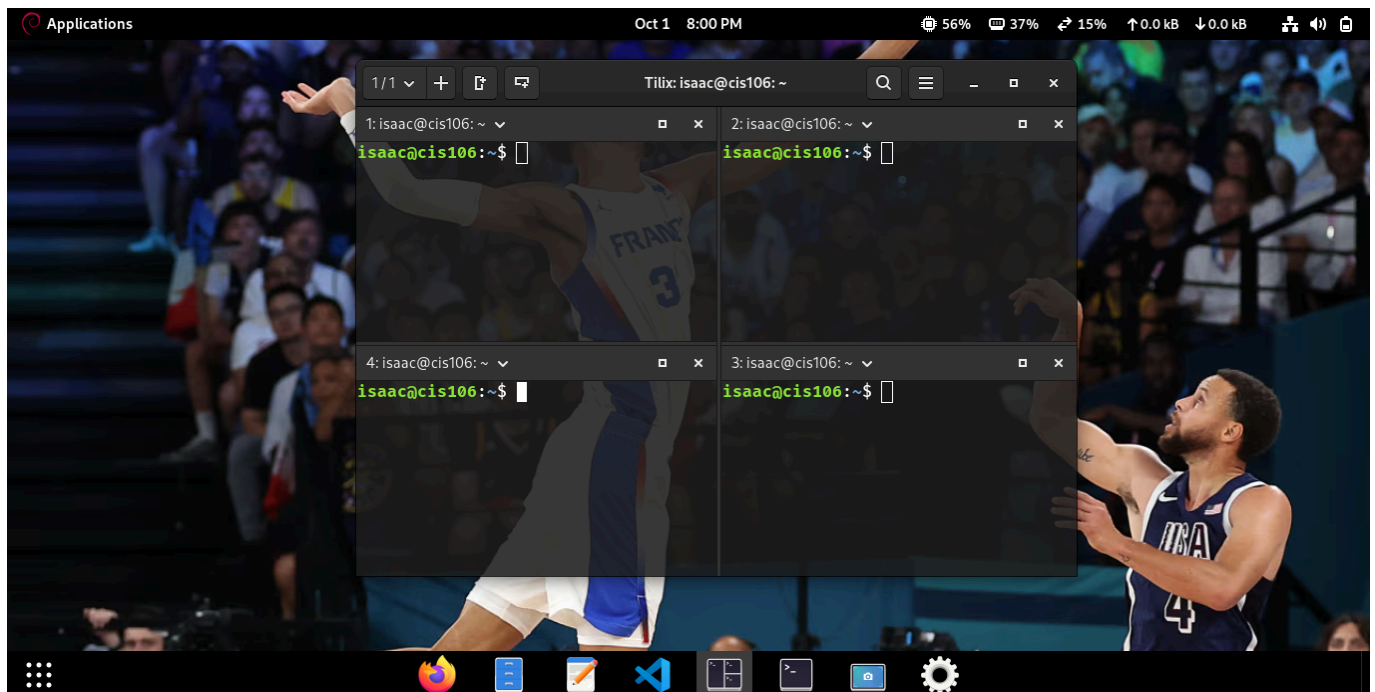


Lab 3 Submission

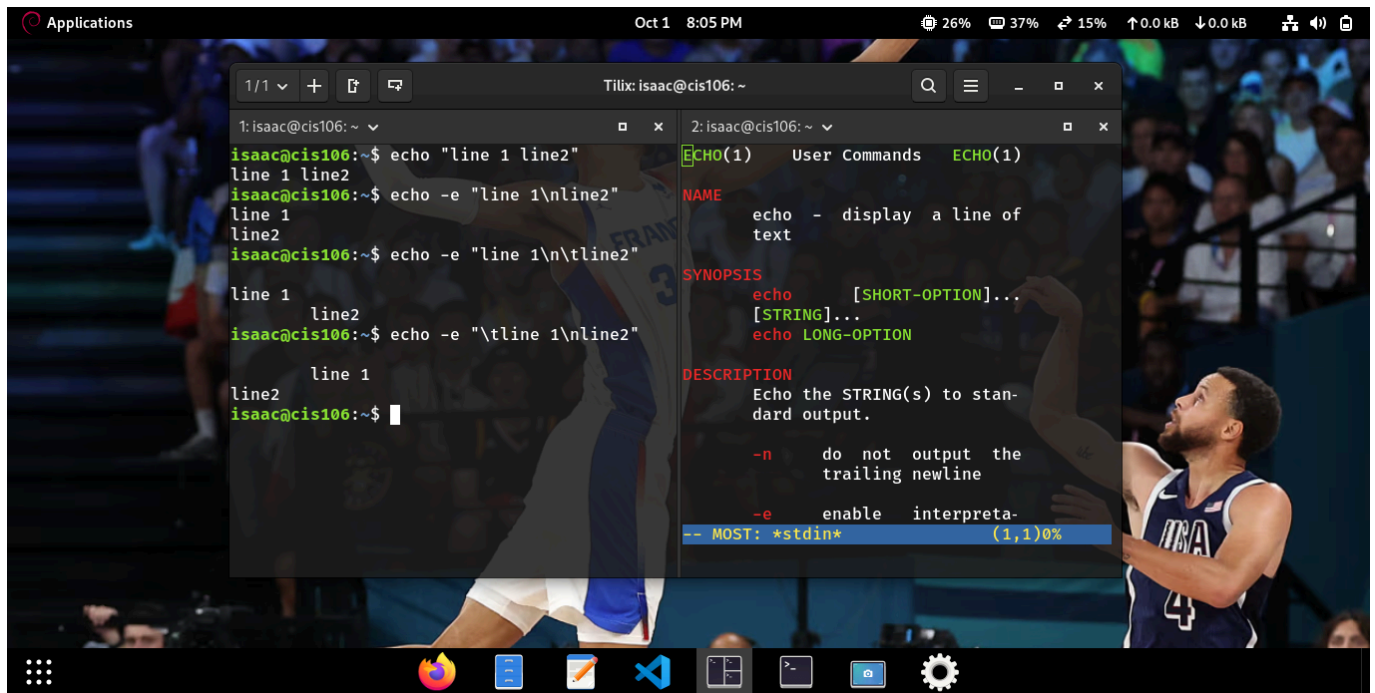
Question 1



Question 2



Question 3



The screenshot shows a Linux desktop environment. At the top, the system status bar displays 'Applications', 'Oct 1 8:05 PM', and various resource usage metrics (26% CPU, 37% memory, 15% disk I/O, 0.0 kB network). The desktop background features a basketball player in a blue jersey. A terminal window titled 'Tilix: isaac@cis106: ~' is open, showing a series of echo commands and their outputs. The first command is 'echo "line 1 line2"', which outputs 'line 1 line2'. The second command is 'echo -e "line 1\nline2"', which outputs 'line 1' followed by 'line2' on a new line. The third command is 'echo -e "line 1\n\tline2"', which outputs 'line 1' followed by 'line2' on a new line, with a tab character before 'line2'. The fourth command is 'echo -e "\tline 1\nline2"', which outputs 'line 1' followed by 'line2' on a new line, with a tab character before 'line 1'. To the right of the terminal, a help page for the 'echo' command is displayed, showing its name, synopsis, description, and options. The synopsis is 'echo [SHORT-OPTION]... [STRING]...', where [SHORT-OPTION] can be -n or -e, and [STRING] can be LONG-OPTION. The description states that echo outputs the string(s) to standard output. The options are -n (do not output the trailing newline) and -e (enable interpretation of backslash escapes). The last line of the help page is '-- MOST: *stdin* (1,1)0%'. The desktop environment includes a taskbar at the bottom with icons for Firefox, a file manager, a terminal, and a settings gear.

```
1: isaac@cis106: ~  
isaac@cis106:~$ echo "line 1 line2"  
line 1 line2  
isaac@cis106:~$ echo -e "line 1\nline2"  
line 1  
line2  
isaac@cis106:~$ echo -e "line 1\n\tline2"  
line 1  
line2  
isaac@cis106:~$ echo -e "\tline 1\nline2"  
line 1  
line2  
isaac@cis106:~$
```

ECHO(1) User Commands ECHO(1)

NAME
echo - display a line of text

SYNOPSIS
echo [SHORT-OPTION]... [STRING]...
echo LONG-OPTION

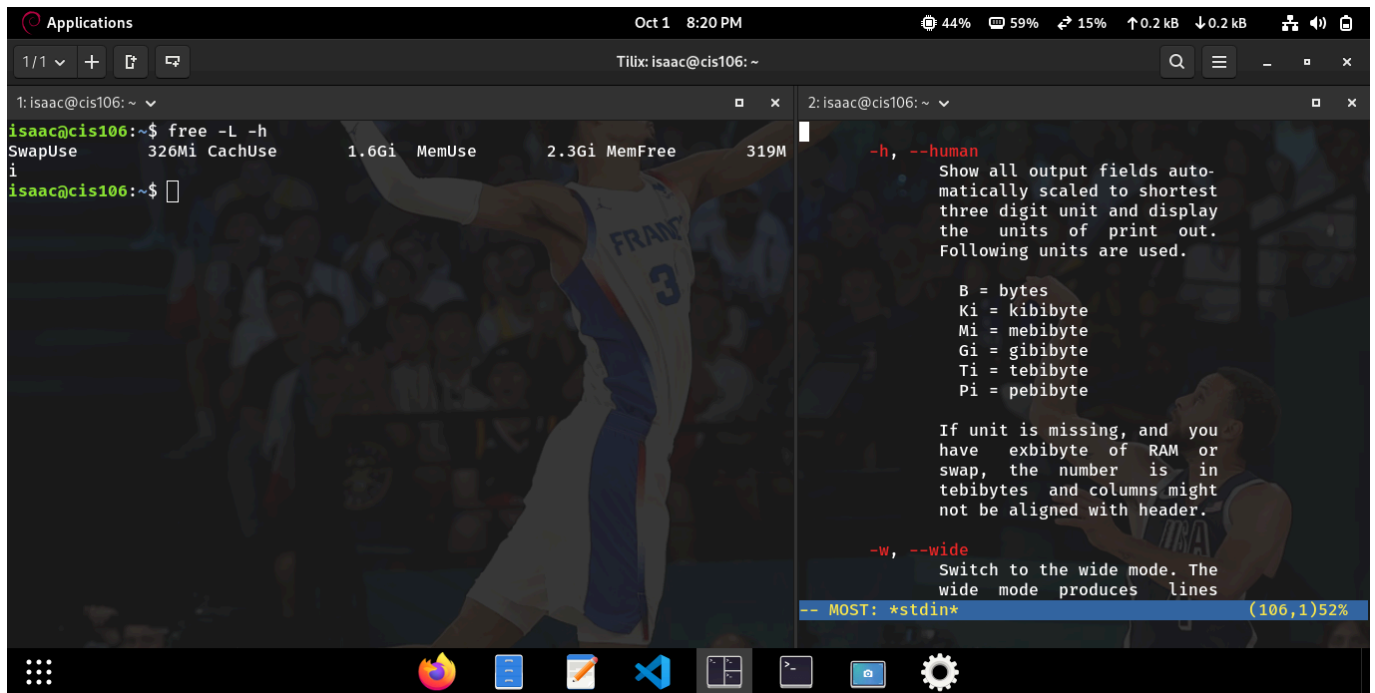
DESCRIPTION
Echo the STRING(s) to standard output.

-n do not output the trailing newline

-e enable interpretation of backslash escapes

-- MOST: *stdin* (1,1)0%

Challenge Questions



```
Applications Oct 1 8:20 PM 44% 59% 15% 0.2 kB 0.2 kB 1/1 + [search] [menu] [close]
Tilix: isaac@cis106: ~
1: isaac@cis106: ~
isaac@cis106:~$ free -L -h
SwapUse      326Mi CachUse      1.6Gi MemUse      2.3Gi MemFree      319M
i
isaac@cis106:~$
```

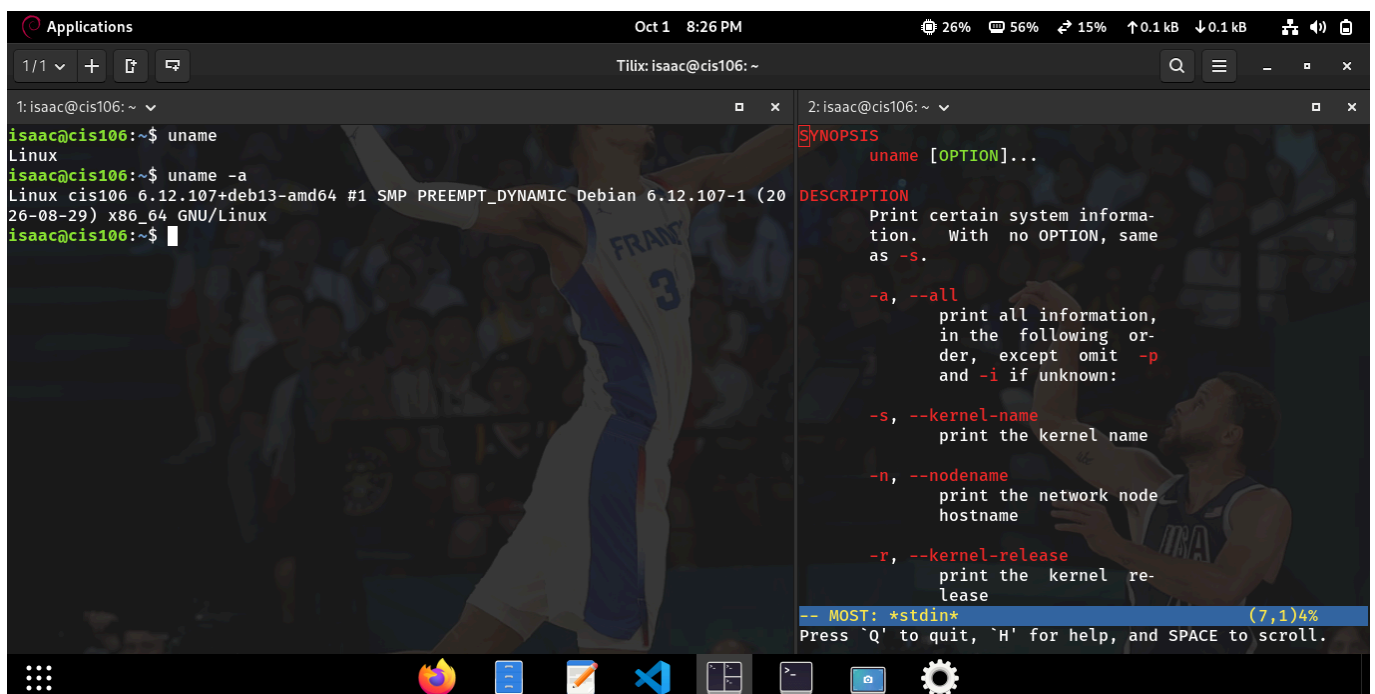
-h, --human
Show all output fields automatically scaled to shortest three digit unit and display the units of print out. Following units are used.

B = bytes
Ki = kibibyte
Mi = mebibyte
Gi = gibibyte
Ti = tebibyte
Pi = pebibyte

If unit is missing, and you have exbibyte of RAM or swap, the number is in tebibytes and columns might not be aligned with header.

-w, --wide
Switch to the wide mode. The wide mode produces lines

-- MOST: *stdin* (106,1)52%



```
Applications Oct 1 8:26 PM 26% 56% 15% 0.1 kB 0.1 kB 1/1 + [search] [menu] [close]
Tilix: isaac@cis106: ~
1: isaac@cis106: ~
isaac@cis106:~$ uname
Linux
isaac@cis106:~$ uname -a
Linux cis106 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64 GNU/Linux
isaac@cis106:~$
```

SYNOPSIS
uname [OPTION]...

DESCRIPTION
Print certain system information. With no OPTION, same as -s.

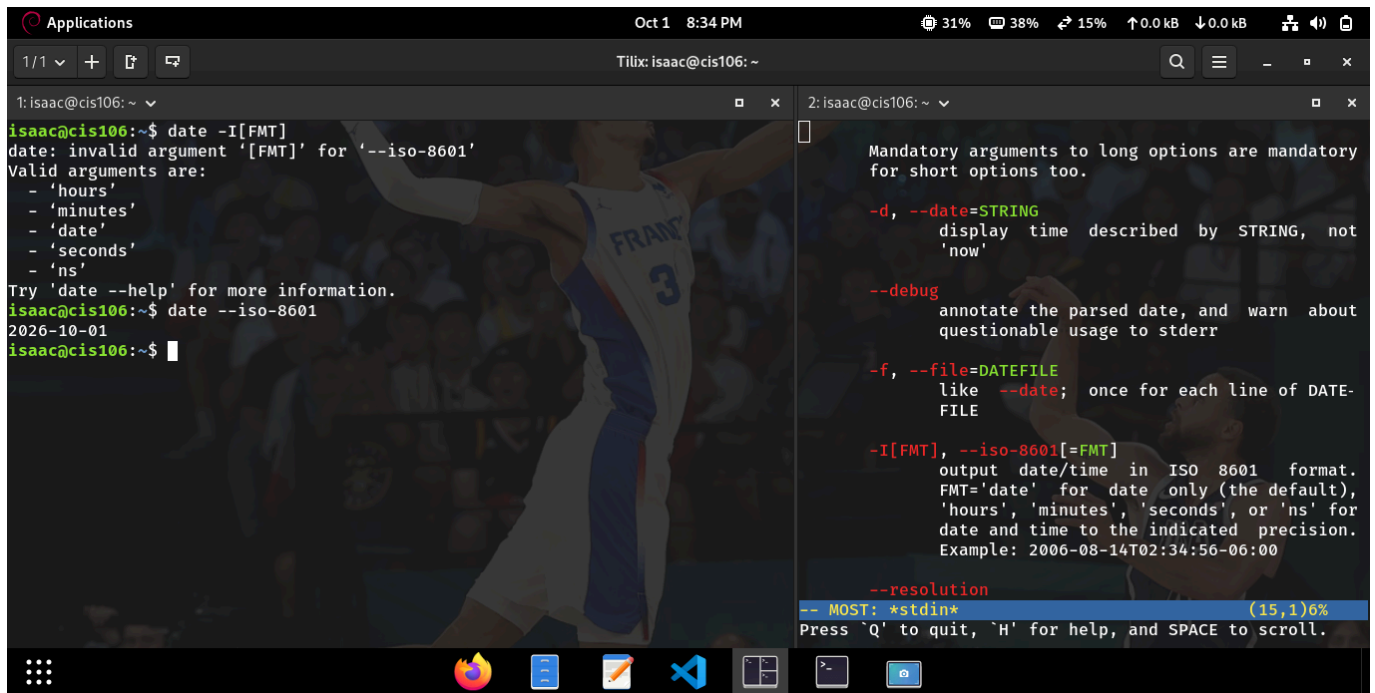
-a, --all
print all information, in the following order, except omit -p and -i if unknown:

-s, --kernel-name
print the kernel name

-n, --nodename
print the network node hostname

-r, --kernel-release
print the kernel release

-- MOST: *stdin* (7,1)4%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.



```
Applications Oct 1 8:34 PM 31% 38% 15% 0.0 kB 0.0 kB
1/1 + [ ]
Tilix: isaac@cis106: ~
1: isaac@cis106: ~
isaac@cis106:~$ date -I[FMT]
date: invalid argument '[FMT]' for '--iso-8601'
Valid arguments are:
- 'hours'
- 'minutes'
- 'date'
- 'seconds'
- 'ns'
Try 'date --help' for more information.
isaac@cis106:~$ date --iso-8601
2026-10-01
isaac@cis106:~$

2: isaac@cis106: ~
Mandatory arguments to long options are mandatory
for short options too.

-d, --date=STRING
    display time described by STRING, not
    'now'

--debug
    annotate the parsed date, and warn about
    questionable usage to stderr

-f, --file=DATEFILE
    like --date; once for each line of DATE-
    FILE

-I[FMT], --iso-8601[=FMT]
    output date/time in ISO 8601 format.
    FMT='date' for date only (the default),
    'hours', 'minutes', 'seconds', or 'ns' for
    date and time to the indicated precision.
    Example: 2006-08-14T02:34:56-06:00

--resolution
-- MOST: *stdin* (15,1)6%
Press 'Q' to quit, 'H' for help, and SPACE to scroll.
```